
1. What is CSS?

CSS stands for Cascading Style Sheets.

It is used to style and layout HTML elements.

CSS controls colors, fonts, spacing, positioning, and responsiveness.

2. Types of CSS

Inline CSS

```
<p style="color: blue;">This is blue text</p>
```

Internal CSS

```
<head>
  <style>
    p {
      color: red;
    }
  </style>
</head>
```

External CSS

```
<link rel="stylesheet" href="styles.css" />
```

Best practice: use external CSS for maintainability.

3. CSS Syntax

```
selector {
  property: value;
}
```

Example:

```
h1 {
  color: green;
  font-size: 32px;
}
```

4. Selectors in CSS

Basic Selectors

* — Universal selector

element — Selects all elements of that type (p, div, etc.)

.class — Selects elements with a class

#id — Selects a single element with a specific id

Combinators

div p — Descendant

div > p — Direct child

div + p — Adjacent sibling

5. Colors in CSS

color: red;

color: #ff0000;

color: rgb(255, 0, 0);

You can use:

Color names

HEX codes

RGB / RGBA

HSL / HSLA

6. Fonts and Text

font-family: Arial, sans-serif;

font-size: 18px;

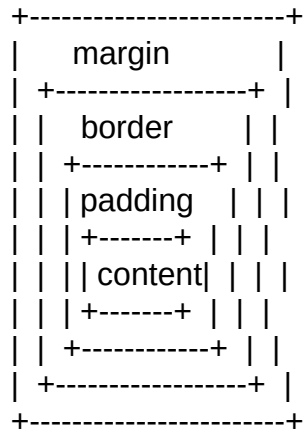
font-weight: bold;

text-align: center;

```
text-decoration: underline;
text-transform: uppercase;
```

7. Box Model

Every element is a box with:



CSS Example:

```
div {
  margin: 20px;
  padding: 10px;
  border: 2px solid black;
}
```

8. Layout (Display & Position)

Display

display: block | inline | inline-block | flex | grid | none;

Position

```
position: static | relative | absolute | fixed | sticky;
top: 10px;
left: 20px;
```

9. Flexbox (Responsive Layout)

```
.container {
  display: flex;
  justify-content: space-between;
  align-items: center;
```

```
}
```

Common Properties:

flex-direction

justify-content

align-items

flex-wrap

10. Responsive Design

Media Queries

```
@media screen and (max-width: 768px) {  
  body {  
    background-color: lightgray;  
  }  
}
```

11. Transitions and Animations

Transition

```
button {  
  transition: background-color 0.3s;  
}
```

Animation

```
@keyframes slide {  
  from { left: 0; }  
  to { left: 100px; }  
}
```

```
div {  
  animation: slide 1s ease-in-out;  
}
```
